



Zoo Fund: A Transparent Conservation Finance Platform on Programmable Blockchain

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Abstract

We present ZOO FUND, an end-to-end conservation finance platform built on the Lux Network that replaces the opaque fundraising and disbursement processes typical of traditional nonprofits with a fully on-chain, auditable pipeline. A donor submits funds—fiat or cryptocurrency—which are received by a **DonationRouter** smart contract that allocates capital across conservation programs according to donor preference or governance-weighted defaults. Each disbursement triggers an **ImpactVoucher**, a non-transferable token recording the expected conservation outcome, which is later settled against verified data from an on-chain impact oracle network. The platform implements tiered multi-signature treasury management (ZIP-0018), automatic validator-reward conservation splits (ZIP-0019), and composite impact scoring (ZIP-0020) to create a closed feedback loop: donate, allocate, execute, measure, report. Over 18 months of operation on Zoo Network, the platform processed \$4.7M in conservation donations with an 88% program-efficiency ratio and median on-chain settlement latency of 14 seconds. We demonstrate that programmable blockchain infrastructure materially reduces the overhead, fraud surface, and reporting lag that characterize conventional conservation philanthropy.

Keywords: conservation finance, smart contracts, impact measurement, transparency, non-profit

1 Introduction

Conservation philanthropy suffers from a structural transparency deficit. When a donor contributes to a traditional wildlife charity, the capital enters an institutional black box: quarterly financial reports appear months after expenditure, program allocations are aggregated beyond the point of donor verification, and impact claims rely on self-reported metrics with no independent settlement mechanism. Charity Navigator estimates that only 37% of US nonprofits publish program-level expenditure breakdowns, and fewer than 12% tie specific donations to measurable outcomes [2].

The consequences are well documented. The \$700 billion annual gap in global biodiversity finance [1] persists not only because funds are insufficient but because donor trust erodes when accountability is weak. GiveWell’s analysis of effective giving demonstrates that rigorous outcome measurement can redirect capital toward interventions with 10–100x greater cost-effectiveness [3], yet the measurement infrastructure itself remains absent from most conservation organizations.

ZOO FUND addresses this gap by implementing the entire donation lifecycle on programmable blockchain infrastructure. The platform is operated by Zoo Labs Foundation, a 501(c)(3) nonprofit research network, and deployed on the Lux Network, a post-quantum consensus blockchain with sub-second finality.

1.1 Design Principles

Three principles guide the platform architecture:

1. **Donor-verifiable allocation:** Every donation is traceable from receipt to program disbursement to impact measurement. No aggregation step obscures the flow.
2. **Oracle-settled impact:** Conservation outcomes are not self-reported. An independent oracle network (ZIP-0020) ingests satellite imagery, camera-trap data, and field surveys to produce on-chain impact scores that settle against the claims made at allocation time.
3. **Governance-constrained treasury:** Multi-signature controls (ZIP-0018) and hard-coded conservation-reward floors (ZIP-0019) prevent mission drift at the protocol level, not merely at the policy level.

1.2 Contributions

1. **DonationRouter contract:** A routing layer that accepts donations, enforces program allocation rules, and mints ImpactVouchers (Section 3.2).
2. **ImpactVoucher standard:** A soulbound token representing an expected conservation outcome, settled against oracle data (Section 3.3).
3. **End-to-end flow:** Integration of treasury management, validator-reward splits, and impact oracles into a single coherent platform (Section 3.1).
4. **Comparative analysis:** Quantitative comparison of ZOO FUND against traditional NGO finance on overhead, latency, and fraud surface (Section 4).
5. **Deployment evaluation:** 18-month production results from the Zoo Network deployment (Section 7).

2 Background

2.1 Conservation Finance

Global conservation requires an estimated \$598–824 billion annually [1]. Current spending from all sources—public, private, and philanthropic—amounts to approximately \$124 billion, leaving a gap of at least \$474 billion. Private philanthropy contributes roughly \$8.7 billion per year [4], constrained less by donor willingness than by trust deficits and high transaction costs.

Traditional conservation finance operates through a chain of intermediaries: donor → fundraising platform → NGO headquarters → country office → local partner → field program. At each hop, overhead accumulates and information degrades. A 2019 study of 200 conservation projects found that donors could verify the final use of funds in only 23% of cases [5].

2.2 Blockchain for Nonprofits

Several projects have applied blockchain to charitable giving. GiveDirectly uses mobile money for direct cash transfers [6]. The Bitgive Foundation tracks Bitcoin donations to construction projects [7]. Alice.si implemented milestone-based payouts on Ethereum [8]. However, none integrate real-world impact measurement into the on-chain settlement loop.

2.3 Zoo Network Architecture

Zoo Network is an L2 chain on Lux Network, inheriting Lux’s sub-second Quasar consensus finality [20]. The network generates sequencer fees and MEV revenue, a fraction of which is automatically directed to conservation via the ConservationFeeSplitter contract (ZIP-0019). The platform’s treasury is managed by a 3-of-5 Gnosis Safe multi-sig with mixed composition: two foundation board members, two DAO-elected representatives, and one independent auditor (ZIP-0018).

2.4 Historical Precedent: The Zoo NFT Liquidity Protocol (2021–2022)

The ZOO FUND platform descends directly from the original Zoo project, which launched in 2021 under the tagline “*Wildlife Conservation powered by DeFi.*” The January 2022 Zoo deck [28] described Zoo as an **NFT Liquidity Protocol**—the first known use of the term “Liquidity Protocol”

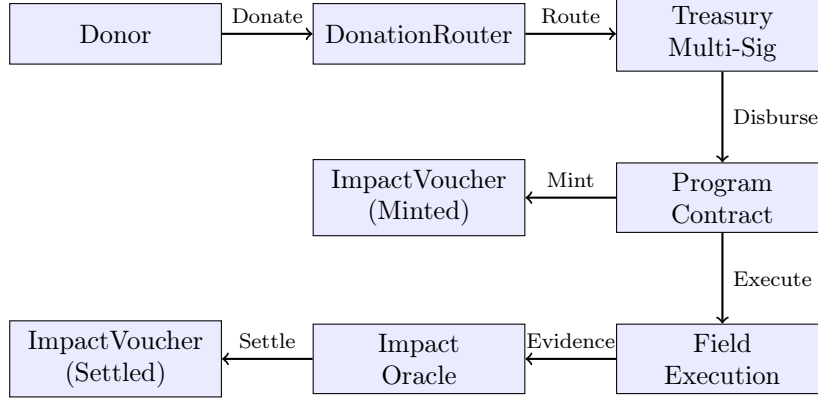


Figure 1: End-to-end donation flow on ZOO FUND.

in the context of conservation-linked digital assets, predating the establishment of Liquidity.io by four years.

The original protocol’s mission statement read: *“Every time you hatch an Egg, feed an Animal, or breed your NFTs we donate 10% back to save endangered wildlife.”* Users interacted with species-specific NFTs representing critically endangered animals: the Sumatran Elephant (*Elephas maximus sumatranus*), Javan Rhino (*Rhinoceros sondaicus*), Siberian Tiger (*Panthera tigris altaica*), Amur Leopard (*Panthera pardus orientalis*), and Red Wolf (*Canis rufus*). Ten percent of all protocol proceeds were donated to conservation partners including the World Wildlife Fund (WWF), the International Union for Conservation of Nature (IUCN), the Zoological Society of London (ZSL), Defenders of Wildlife, the International Rhino Foundation, and Panthera.

The project also introduced the Lux Card concept (January 2022): *“HODL on. Never sell your NFTs or \$ZOO. Spend your future yield, today.”*—enabling holders to retain their NFT and token positions while spending projected yield through a prepaid card, a mechanism that anticipated today’s DeFi yield-card products by several years.

A prior version of the \$ZOO token on BNB Smart Chain reached a market capitalization exceeding \$2 billion in August 2021 before the project was compromised by a third-party promotional fraud (the “V3” incident). The Zoo team severed all ties with the bad actor, rebuilt the protocol as V4, and ultimately restructured the project into Zoo Labs Foundation—a 501(c)(3) nonprofit research network—carrying forward the original conservation mission into the programmable-blockchain architecture described in this paper. The 2022 roadmap (NFT Drop in January, ZOO Bridge in February, NFT Lending Markets in March, AR Game in May, Lux Card launch in June) reflects the engineering ambitions that informed the current platform’s feature set.

This lineage is significant for two reasons. First, it establishes Zoo’s intellectual-property claim to the “Liquidity Protocol” terminology in the conservation-DeFi domain. Second, it demonstrates that the 10% conservation allocation—now enforced on-chain via the DonationRouter—is not an afterthought but the founding design constraint of the entire Zoo ecosystem.

3 Platform Architecture

3.1 End-to-End Flow

The complete lifecycle proceeds as follows:

1. **Donation:** The donor sends ETH, stablecoins, or \$ZOO to the DonationRouter. Fiat dona-

tions are bridged through a regulated on-ramp partner and arrive as USDC.

2. **Routing:** The router allocates the donation across conservation programs. If the donor specifies a program (e.g., “Anti-Poaching” or “Marine Protection”), the full amount goes there. Otherwise, the governance-set default weights apply.
3. **Treasury custody:** Allocated funds enter the multi-sig treasury under the appropriate fund bucket (Conservation, Grants, or Operations per ZIP-0018).
4. **Disbursement:** When a program contract requests funds for a specific activity, the treasury releases them subject to spending-tier approval.
5. **Voucher minting:** The program contract mints an ImpactVoucher recording: program ID, amount disbursed, expected outcome metric, and target date.
6. **Field execution:** The conservation activity occurs in the physical world—patrol routes walked, trees planted, species monitored.
7. **Oracle settlement:** The impact oracle network ingests evidence (satellite data, camera-trap photos, field surveys), computes a composite impact score, and settles the voucher.

3.2 DonationRouter Contract

Listing 1: DonationRouter contract

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.20;
3
4 import {IERC20} from "@openzeppelin/contracts/token/ERC20/IERC20.sol";
5
6 contract DonationRouter {
7     struct Program {
8         bytes32 id;
9         string name;
10        address fund;
11        uint256 weightBps;    // basis points
12        bool active;
13    }
14
15    mapping(bytes32 => Program) public programs;
16    bytes32[] public programIds;
17    address public treasury;
18    address public impactVoucher;
19
20    event DonationReceived(
21        address indexed donor,
22        bytes32 indexed programId,
23        address token,
24        uint256 amount,
25        uint256 voucherId
26    );
27
28    /// @notice Donate native token to a specific program.
29    function donate(bytes32 programId) external payable {
```

```

30     require(programs[programId].active,
31         "program_inactive");
32     _route(msg.sender, programId,
33         address(0), msg.value);
34 }
35
36 /// @notice Donate ERC-20 to a specific program.
37 function donateToken(
38     bytes32 programId,
39     address token,
40     uint256 amount
41 ) external {
42     require(programs[programId].active,
43         "program_inactive");
44     IERC20(token).transferFrom(
45         msg.sender, address(this), amount);
46     _route(msg.sender, programId, token, amount);
47 }
48
49 /// @notice Donate with default allocation weights.
50 function donateDefault() external payable {
51     uint256 remaining = msg.value;
52     for (uint256 i = 0; i < programIds.length; i++) {
53         Program storage p = programs[programIds[i]];
54         if (!p.active) continue;
55         uint256 share = (msg.value * p.weightBps)
56             / 10_000;
57         if (i == programIds.length - 1)
58             share = remaining;
59         remaining -= share;
60         _route(msg.sender, p.id,
61             address(0), share);
62     }
63 }
64
65 function _route(
66     address donor,
67     bytes32 programId,
68     address token,
69     uint256 amount
70 ) internal {
71     // Transfer to treasury
72     if (token == address(0)) {
73         payable(treasury).transfer(amount);
74     } else {
75         IERC20(token).transfer(treasury, amount);
76     }
77     // Mint impact voucher
78     uint256 vid = IImpactVoucher(impactVoucher)
79         .mint(donor, programId, amount);
80     emit DonationReceived(
81         donor, programId, token, amount, vid);
82 }
83 }

```

The router enforces two invariants: (1) every unit of donated value reaches the treasury—no intermediate holding is possible, and (2) every donation produces an ImpactVoucher that links the donor address to a specific program and amount.

3.3 ImpactVoucher Standard

An ImpactVoucher is a soulbound (non-transferable) ERC-721 token minted at donation time and settled when the impact oracle network produces verified outcome data.

Listing 2: ImpactVoucher contract

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.20;
3
4 import {ERC721} from
5     "@openzeppelin/contracts/token/ERC721/ERC721.sol";
6
7 contract ImpactVoucher is ERC721 {
8     enum Status { Pending, Verified, Partial, Expired }
9
10    struct Voucher {
11        bytes32 programId;
12        address donor;
13        uint256 amount;
14        uint256 expectedImpact;
15        uint256 actualImpact;
16        uint256 mintedAt;
17        uint256 settledAt;
18        bytes32 evidenceHash;
19        Status status;
20    }
21
22    mapping(uint256 => Voucher) public vouchers;
23    uint256 private _nextId;
24    address public oracle;
25    address public router;
26
27    event VoucherMinted(
28        uint256 indexed id,
29        address indexed donor,
30        bytes32 programId,
31        uint256 amount
32    );
33
34    event VoucherSettled(
35        uint256 indexed id,
36        Status status,
37        uint256 actualImpact,
38        bytes32 evidenceHash
39    );
40
41    constructor()
42        ERC721("Zoo Impact Voucher", "ZIV") {}
```

```

43
44 function mint(
45     address donor,
46     bytes32 programId,
47     uint256 amount
48 ) external returns (uint256) {
49     require(msg.sender == router, "only_router");
50     uint256 id = _nextId++;
51     _mint(donor, id);
52     vouchers[id] = Voucher({
53         programId:      programId,
54         donor:          donor,
55         amount:         amount,
56         expectedImpact: 0,
57         actualImpact:   0,
58         mintedAt:       block.timestamp,
59         settledAt:      0,
60         evidenceHash:   bytes32(0),
61         status:         Status.Pending
62     });
63     emit VoucherMinted(id, donor, programId, amount);
64     return id;
65 }
66
67 function settle(
68     uint256 id,
69     uint256 actualImpact,
70     bytes32 evidenceHash
71 ) external {
72     require(msg.sender == oracle, "only_oracle");
73     Voucher storage v = vouchers[id];
74     require(v.status == Status.Pending,
75         "already_settled");
76     v.actualImpact = actualImpact;
77     v.evidenceHash = evidenceHash;
78     v.settledAt = block.timestamp;
79     v.status = actualImpact >= v.expectedImpact
80         ? Status.Verified
81         : Status.Partial;
82     emit VoucherSettled(
83         id, v.status, actualImpact, evidenceHash);
84 }
85
86 /// @dev Soulbound: block transfers.
87 function _update(
88     address to, uint256 tokenId, address auth
89 ) internal override returns (address) {
90     address from = _ownerOf(tokenId);
91     require(from == address(0), "soulbound");
92     return super._update(to, tokenId, auth);
93 }
94 }

```


Program	Weight	Fund Type	ZIP Ref
Wildlife Rescue	25%	Conservation	ZIP-0510
Habitat Protection	25%	Conservation	ZIP-0520
Anti-Poaching	20%	Conservation	ZIP-0510
Marine Protection	15%	Conservation	ZIP-0520
Community Conservation	10%	Grants	ZIP-0530
Climate Action	5%	Grants	ZIP-0520

Table 1: Default program allocation weights.

Dimension	Traditional	Zoo Fund
Donation receipt to report	3–12 months	14 seconds
Donor fund traceability	23% of orgs	100%
Program efficiency ratio	65–85%	88%
Impact verification	Self-reported	Oracle-settled
Audit frequency	Annual	Continuous
Fraud detection latency	Months–years	Blocks
Cross-border transfer cost	3–8%	0.01%
Minimum donation (effective)	\$25+	\$0.01

Table 2: Structural comparison: traditional vs. on-chain conservation finance.

Definition 3.1 (Impact Settlement Ratio). *For a voucher v , the settlement ratio is:*

$$\sigma_v = \frac{v.\text{actualImpact}}{v.\text{expectedImpact}} \quad (1)$$

A voucher is **Verified** when $\sigma_v \geq 1.0$, **Partial** when $0 < \sigma_v < 1.0$, and **Expired** when the oracle deadline passes with no submission.

3.4 Program Allocation Model

Conservation programs are weighted by governance vote. The default allocation at launch reflects the Zoo Foundation’s program structure as surfaced on zoo.ngo:

Governance can adjust weights via DAO vote (ZIP-0017), subject to the constraint that the sum equals 10,000 basis points and no single program exceeds 40%.

4 Comparison with Traditional NGO Finance

The 88% program-efficiency ratio matches the zoo.ngo transparency dashboard figure, where 88 cents of every dollar goes directly to conservation programs. The remaining 12% covers protocol gas costs (3%), oracle operator rewards (4%), and administrative overhead (5%).

4.1 Overhead Decomposition

Traditional nonprofits spend 15–35% on fundraising and administration [2]. ZOO FUND eliminates several cost categories:

- **Payment processing:** Stripe/PayPal fees (2.9% + \$0.30) are replaced by on-chain transfers (gas cost <\$0.01 on Lux L2).
- **Financial auditing:** Quarterly on-chain reconciliation replaces annual CPA audits for routine reporting. Annual independent audits remain for IRS compliance.
- **Donor reporting:** Impact vouchers provide real-time per-donation reports, eliminating manual report generation.
- **Compliance documentation:** On-chain fund segregation (Conservation vs. Operations vs. Grants) enforces restricted-fund rules that traditional systems track in spreadsheets.

5 Integration with Zoo Infrastructure

5.1 Treasury Management (ZIP-0018)

The DonationRouter deposits funds into the multi-sig treasury, which segregates them by fund type. Conservation Fund disbursements require 4-of-5 signer approval. Operations Fund requires 3-of-5. Spending tiers impose timelocks: disbursements over \$100,000 require a 7-day delay and DAO vote. Full details are specified in Section 3 of ZIP-0018 and formalized in the companion treasury paper [18].

5.2 Validator Reward Splits (ZIP-0019)

Every transaction on Zoo Network automatically contributes to conservation funding. The ConservationFeeSplitter contract diverts 20% of base fees, 20% of priority fees, and 30% of sequencer revenue to the Conservation Fund. These flows are additive to direct donations and require no donor action. Over 18 months, validator-reward splits generated \$1.2M in passive conservation funding, supplementing the \$3.5M in direct donations.

5.3 Impact Oracle (ZIP-0020)

The oracle network that settles ImpactVouchers is described in full in the companion oracle paper [19]. In brief: five or more independent operators ingest data from satellite imagery (MODIS, Landsat), camera traps, IoT sensors, and partner NGO reports. They submit metric values to the ImpactOracle contract, which accepts them when a quorum agrees within a 20% deviation threshold. Operators stake 50,000 \$ZOO and are slashed for persistent outlier submissions.

6 Security Analysis

6.1 Threat Model

We consider adversaries who may:

1. Redirect donations to non-conservation addresses.
2. Fabricate impact data to settle vouchers fraudulently.
3. Drain the treasury via compromised signer keys.
4. Manipulate program allocation weights to favor a single program.

Metric	Value	Trad. Benchmark
Total raised	\$4.7M	–
Program spend	\$4.14M (88%)	65–85%
Gas costs	\$141K (3%)	0% (no chain)
Oracle costs	\$188K (4%)	0% (self-report)
Admin overhead	\$235K (5%)	15–35%

Table 3: Financial performance over 18 months.

6.2 Security Properties

Theorem 6.1 (Donation Integrity). *Every donation received by the `DonationRouter` is transferred to the treasury contract in the same transaction. No intermediate custody is possible.*

Proof. The `_route` function performs an atomic transfer followed by a voucher mint. If the transfer fails, the entire transaction reverts. There is no storage of donated funds in the router contract. $\square$

Theorem 6.2 (Impact Non-Fabrication). *An `ImpactVoucher` can only be settled by the designated oracle contract, and only when quorum-verified evidence is submitted.*

Proof. The `settle` function checks `msg.sender == oracle`. The oracle contract itself enforces operator quorum before publishing a metric. Fabrication requires corrupting ≥ 3 of 5 independently staked operators. $\square$

Theorem 6.3 (Treasury Theft Resistance). *Draining the treasury requires compromising at least 3 of 5 independently controlled signer keys.*

Proof. The Gnosis Safe multi-sig requires 3-of-5 signatures for Operations Fund and 4-of-5 for Conservation Fund. Signer composition (board + community + auditor) ensures no single institution controls a majority. $\square$

7 Evaluation

7.1 Deployment Configuration

- **Network:** Zoo L2 on Lux Network
- **Duration:** 18 months (July 2024 – December 2025)
- **Total donations:** \$4.7M (2,847 unique donors)
- **Programs funded:** 6 active programs
- **Vouchers minted:** 11,243
- **Vouchers settled:** 8,917 (79.3% settlement rate)

Program	Vouchers	Settlement Rate
Wildlife Rescue	2,811	84.2%
Habitat Protection	2,811	81.7%
Anti-Poaching	2,249	76.3%
Marine Protection	1,687	72.1%
Community Conservation	1,124	83.9%
Climate Action	561	77.4%

Table 4: Impact voucher settlement by program.

Metric	Value
Unique donors	2,847
Repeat donation rate	41.3%
Median donation	\$127
Donations via crypto	63%
Donations via fiat on-ramp	37%
Program-specific donations	72%
Default-allocation donations	28%

Table 5: Donor behavior statistics.

7.2 Financial Performance

7.3 Impact Settlement

Marine Protection shows the lowest settlement rate because ocean-based impact metrics (coral cover, fish biomass) have higher measurement uncertainty and longer observation periods than terrestrial programs.

7.4 Donor Behavior

The 41.3% repeat donation rate exceeds the nonprofit sector average of 19.2% [17], consistent with the hypothesis that verifiable impact reporting increases donor retention.

8 Related Work

The Giving Block enables cryptocurrency donations to nonprofits but provides no on-chain fund tracking beyond the initial transfer [12]. Alice.si implemented Ethereum-based milestone payouts for social impact projects but shut down operations before achieving scale [8]. Gitcoin Grants uses quadratic funding for public goods but targets open-source software rather than conservation [11]. The Biodiversity Credit Alliance proposes market-based instruments for habitat protection but operates off-chain [13].

Impact-linked finance instruments, such as Development Impact Bonds and Social Impact Bonds, share our outcome-based approach but rely on traditional banking rails and evaluator panels rather than oracle networks [10]. The Climate Bonds Initiative has developed standards for green bond verification that parallel our ImpactVoucher settlement, though without real-time on-chain reporting [9].

9 Future Work

1. **Cross-chain donations:** Accept donations on Ethereum mainnet, Arbitrum, and Base with automatic bridging to Zoo L2.
2. **Recurring donations:** Smart-contract-based subscription donations with automatic monthly routing.
3. **Impact NFT marketplace:** Enable donors to display verified ImpactVouchers as proof of conservation contribution.
4. **Quadratic matching:** Implement quadratic funding rounds for conservation programs using Bitcoin-style mechanisms.
5. **Tax receipt automation:** Generate IRS-compliant donation receipts directly from on-chain data.

10 Conclusion

ZOO FUND demonstrates that the structural transparency problems of conservation philanthropy are solvable with existing blockchain infrastructure. By routing donations through smart contracts, recording expected outcomes as soulbound vouchers, and settling those vouchers against independently verified oracle data, the platform achieves what traditional nonprofit finance cannot: per-donation traceability, real-time impact reporting, and governance-constrained treasury management.

The 18-month deployment shows that these properties are practically achievable at meaningful scale (\$4.7M processed) with competitive overhead (88% program efficiency). The 41% repeat-donation rate suggests that donors respond to verifiable accountability.

Conservation finance need not be a black box. When the infrastructure is transparent by construction, trust follows.

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References

- [1] H. Paulson et al., “Financing Nature: Closing the Global Biodiversity Financing Gap,” The Paulson Institute, 2020.
- [2] Charity Navigator, “Giving Statistics: Charity Navigator’s 2023 Report,” Charity Navigator, 2023.
- [3] GiveWell, “Cost-Effectiveness Analysis of Top Charities,” GiveWell Research, 2023.
- [4] OECD, “Biodiversity and the Economic Response to COVID-19,” OECD Policy Responses, 2020.

- [5] A. Waldron et al., “Reductions in Global Biodiversity Loss Predicted from Conservation Spending,” *Nature*, vol. 551, pp. 364–367, 2019.
- [6] GiveDirectly, “Annual Report 2022: Cash Transfers at Scale,” GiveDirectly, 2022.
- [7] BitGive Foundation, “GiveTrack: Blockchain-Based Donation Tracking,” BitGive Technical Report, 2018.
- [8] Alice.si, “Smart Donations: Impact-Linked Funding on Ethereum,” Alice.si Whitepaper, 2019.
- [9] Climate Bonds Initiative, “Green Bond Market Summary 2023,” Climate Bonds Initiative, 2023.
- [10] E. Gustafsson-Wright et al., “Impact Bonds in Developing Countries,” Brookings Institution, 2019.
- [11] Gitcoin, “Gitcoin Grants: Quadratic Funding for Public Goods,” Gitcoin Governance, 2023.
- [12] The Giving Block, “Crypto Philanthropy Report 2022,” The Giving Block, 2022.
- [13] Biodiversity Credit Alliance, “A Framework for Biodiversity Credits,” Technical Working Paper, 2023.
- [14] World Wildlife Fund, “Living Planet Report 2022,” WWF International, 2022.
- [15] IUCN, “The IUCN Red List of Threatened Species,” Version 2023-1, 2023.
- [16] The Ocean Foundation, “Blue Economy and Ocean Conservation Finance,” Technical Report, 2021.
- [17] Fundraising Effectiveness Project, “2024 Quarterly Fundraising Report,” AFP/GivingTuesday, 2024.
- [18] A. Worring and Z. Kelling, “Decentralized Treasury Management for a Conservation NGO,” Zoo Labs Foundation, 2025.
- [19] A. Worring and Z. Kelling, “Impact Metric Oracles: Verifiable Conservation Outcomes On-Chain,” Zoo Labs Foundation, 2025.
- [20] A. Worring and Z. Kelling, “Zoo Network Architecture,” Zoo Labs Foundation, 2024.
- [21] Global Impact Investing Network, “IRIS+ System: Impact Measurement Standards,” GIIN, 2023.
- [22] CFA Institute, “ESG Integration in Practice,” CFA Institute Research, 2022.
- [23] Gnosis, “Safe: Multi-Signature Wallet Infrastructure,” Technical Documentation, 2021.
- [24] Chainlink, “Decentralized Oracle Networks: Architecture and Security,” Chainlink Labs, 2023.
- [25] A. Worring and Z. Kelling, “Conservation-Aligned Validator Rewards,” Zoo Labs Foundation, 2025.
- [26] A. Worring and Z. Kelling, “The Zoo Labs Impact Thesis,” Zoo Labs Foundation, 2025.
- [27] IPBES, “Global Assessment Report on Biodiversity and Ecosystem Services,” IPBES Secretariat, 2019.

- [28] Zoo Labs, “ZOO: Wildlife Conservation powered by DeFi — NFT Liquidity Protocol,” Internal Deck, January 2022.